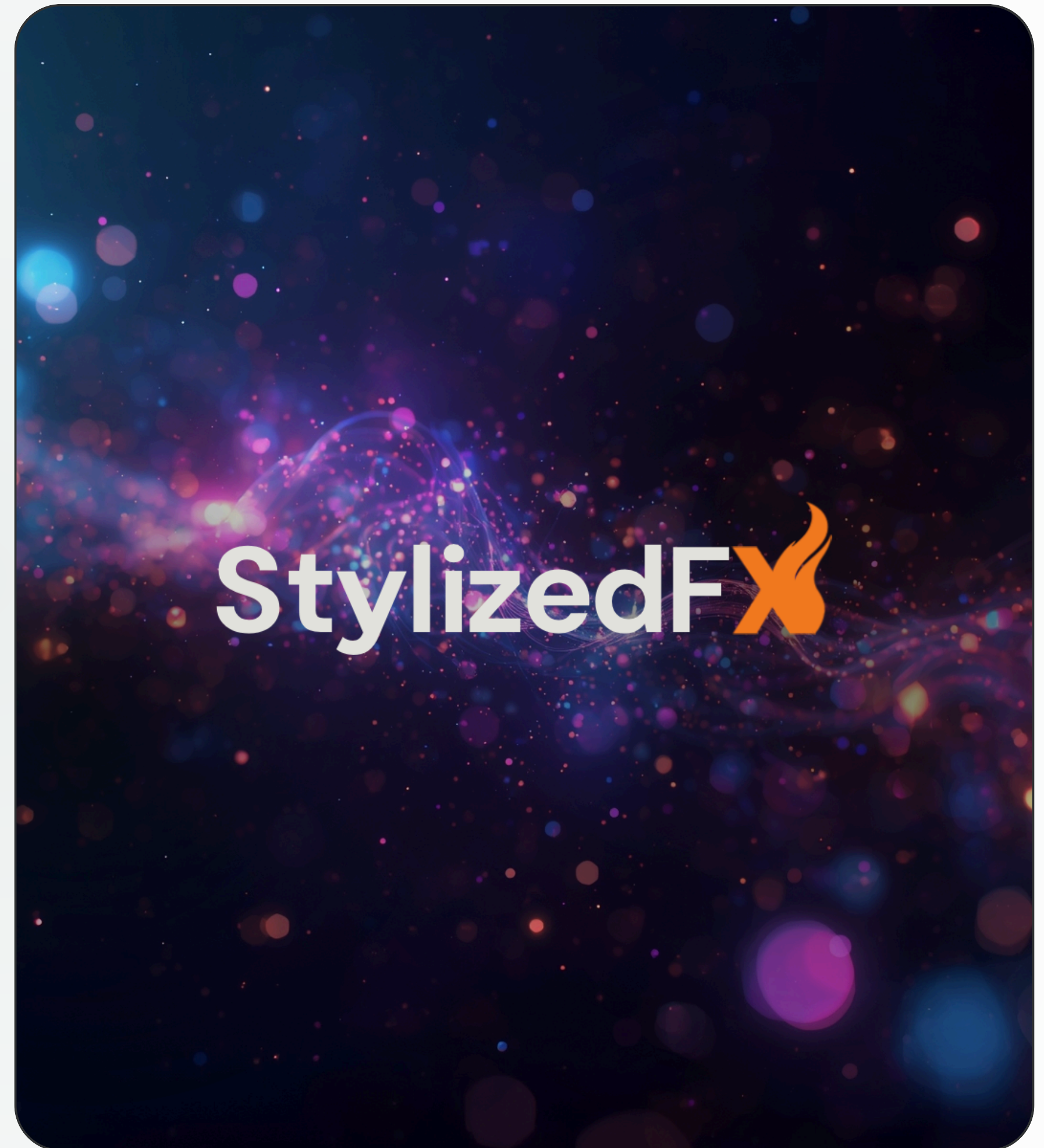


VISUAL EFFECTS

Stylized FX

FAB PROFILE

STYLIZEDFX.COM



VISUAL EFFECTS

StylizedFX

For considering HLSL Shader for your project.

This document contains easy to follow instructions and explanations that will help you get started immediately.

For this product, please write me at stylizedfx@gmail.com



Contents

Unity Version

Unity 6

Shader

HLSL Shader
Universal Render Pipeline (URP)

Platform Compatibility

- iOS (Metal)
- Android (Vulkan / GLES 3.0+)
- Windows / macOS / Linux
- Consoles (Switch / PS / Xbox)

Technical Details

- **Dual Mode:** Includes both Lit (reacts to scene lighting & shadows) and Unlit (pure emissive) variants.
- **Fully Optimized:** Written in efficient HLSL with low ALU count. Perfect for low-end mobile devices to high-end PCs.
- **SRP Batcher Compatible:** CPU-friendly! Render thousands of particles or meshes with minimal draw-call overhead.
- **Distortion & Heat Haze:** Single texture lookup for cost-effective UV distortion effects.
- **Advanced Dissolve:** Smooth erosion effects with customizable edge width, HDR glowing colors, and soft transparency.
- **Soft Particles (Depth Fade):** Eliminates harsh intersections with geometry using optimized Z-buffer comparison.
- **Stylized Lighting:** Built-in "Toon Ramp" support for cel-shaded / anime-style debris and props.
- **Fresnel (Rim Light):** Customizable rim lighting for that magical energy look.
- **Vertex Color Support:** Fully compatible with Unity Particle System's color over lifetime.

Contents

Package Contents:

- VFX_Optimized_Lit.shader (For smoke, debris, toon props)
- VFX_Optimized_Unlit.shader (For lasers, magic, energy orbs)

Demo Materials (Magma, HoloShield, EnergyShell, DarkCore, Toon)

Why use this?

Most asset store shaders are generated by graph editors, resulting in bloated, unreadable code.

This shader is hand-written, clean, and commented, giving you full control and the best possible frame rates.

Contents

Low Mathematical Load (ALU):

- Unity's standard "Lit" or "Complex Lit" shaders rely on Physically Based Rendering (PBR), which involves computationally expensive energy conservation calculations. In contrast, our shader utilizes the Blinn-Phong lighting model—a traditional but highly efficient method. This significantly reduces the processing load on mobile GPUs.

Half Precision:

- We utilize half precision instead of float in the fragment shader. While a float requires 32 bits, a half only requires 16 bits. This optimization reduces memory bandwidth usage and accelerates computation, which is critical for mobile hardware.

Number of Texture Reads:

- Standard PBR shaders typically sample 5–6 textures (e.g., Albedo, Metallic, Roughness, Occlusion, Normal, Emission). Our shader reduces this to just 3 textures (Main, Noise, Bump). Since texture bandwidth is a common bottleneck on mobile devices, this reduction is optimal for performance.

SRP Batcher Compatibility:

- Mobile CPUs often become a performance bottleneck. Thanks to our implementation of CBUFFER macros, the SRP Batcher can now render hundreds of objects (such as bullets or rocks) in a single draw call. This efficiency lowers CPU overhead, reducing heat generation and extending battery life.